Day-44 11 Jun

NAT (Part I). - Static NAT (Source)

Private IPv4 address (RFC 1918):

- IPv4 doesn't provide enough addresses for all devices that need IP address in the modern world.
- The long term solution is to switch to IPv6. But changing networks all over the world from IPv4 to IPv6 ain't simple task :)
- There are three main short-term solutions that've extended life of IPv4. They are as follows:-

  1) CIDR (Classless Inter Domain Routing)
  2) Private IPv4 address
  3) NAT.

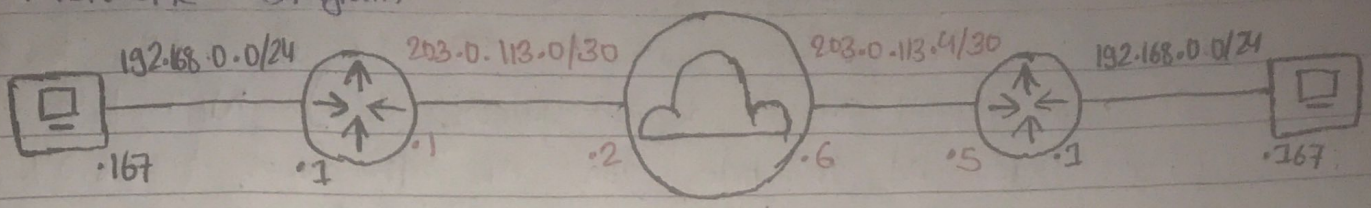
- RFC 1918 specifies the following IPv4 address ranges as private:-
  a) 10.0.0.0/8 (10.0.0.0 to 10.255.255.255) → Class A
  b) 172.16.0.0/12 (172.16.0.0 to 172.31.255.255) → class B
  c) 192.168.0.0/16 (192.168.0.0 to 192.168.255.255) → class C
- We're free to use these addresses in your networks. They don't have to be globally unique.

NOB: Private IP addresses can't be used over the Internet. ISPs will drop traffic to or from private IP addresses.

If private IP add. can't be used over Internet, how can I access internet from PC? That's where NAT comes in.

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Network Diagram



Without NAT, there are two big problems,

i) Duplicate addresses

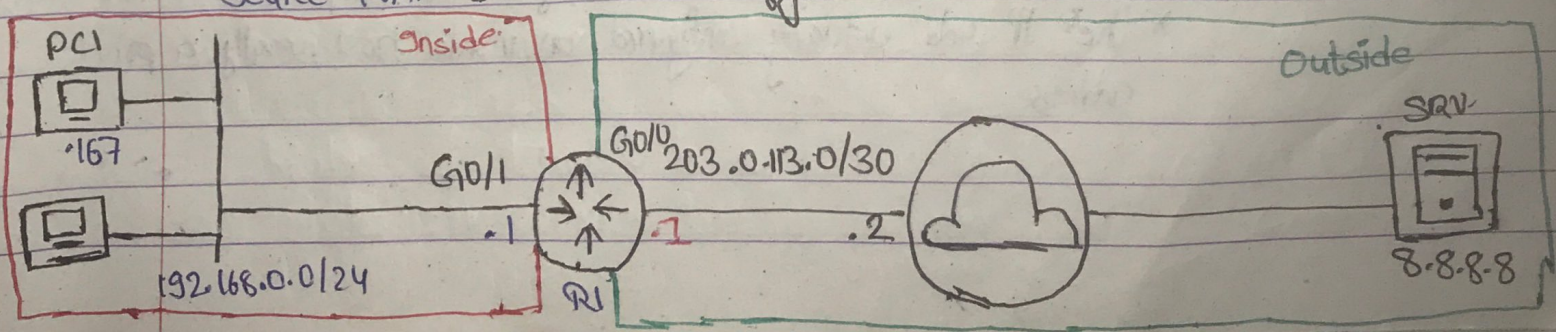
ii) Private IP addresses can't be used over the Internet, so the PCs can't access the Internet.

NAT solves both of this.

Though router's internal interface has private add, its external interface has public address which are unique globally. When our pc (other devices in home network) needs to reach destinations over the Internet, NAT will allow it to borrow the unique public IP add of the router

Network Address Translation (NAT).

- NAT is used to modify the source & and/or destination IP addresses of packets
- There are various reasons to use NAT, but the most ~~import~~ common reason is to allow hosts with private IP addresses to communicate with hosts over the Internet.
- There are various types of NAT, but for CCNA we've to understand 'source NAT' & how to configure it on Cisco routers.



PC1 wants to communicate with SRV at 8.8.8.8. So, it creates packet with source IP 192.168.0.167 & dst 8.8.8.8 & sends packet to its default gateway (R1). NAT takes places here. R1 translates the src IP from 192.168.0.167 to 203.0.113.1 (IP of its external int). That's why its called src NAT. (coz it translates the src IP). The R1 then sends packet out to the internet & it arrives at its destination 8.8.8.8. Now the server will send a reply, where src = 8.8.8.8 & dst = 203.0.113.1. It sends packet to R, which then reverses the translation 203.0.113.1 → 192.168.0.167 Although in this case dst IP is being changed, its not dst NAT. It's just reverting the previously translated IP.

STATIC SOURCE NAT.

- Static NAT involves statistically configuring one-to-one mappings of private IP addresses to public IP addresses.
Note: It doesn't have to be private → public. We can NAT any address.

- An inside local IP is mapped to inside global IP address.
Inside - Router internal network.
Outside - such as internet.

↳ Inside Local - The IP add. of inside host, from the perspective of local
* the* IP add. actually configured on inside host, usually, a priv
address.

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hosts.

↳ Inside global = the IP add. of inside host, from the perspective of outside hosts
* the IP of the inside host after NAT, usually a public address.

NOB* In static src. NAT, we don't use R1 (Router(default_gateway)'s intf IP. there is a separate tt public IP add. reserved for each internal host. eat which is a inside global IP add.

Example:

Say now PC1 wants to communicate with SRV. It sends packet with SRC - 192.168.0.167 & dst 8.8.8.8. Now R1 performs NAT on it & translate it to SRC: 100.0.0.1 & Dst 8.8.8.8. Remember this 100.0.0.1 is public IP reserve mapped to PC1, PC2 can't use it. PC2 will have its own. Now SRV1 receives mssg & sends back reply, with SRC: 8.8.8.8 & dst: 100.0.0.1. R1 then translates the IP from inside global to inside local.

NOB * Static NAT allows devices with private IP add. to communicate over the internet. However because it requires a one-to-one IP add. mapping, it doesn't preserve IP addresses.

So, though there are reasons to use static NAT, it's not the most useful kind of NAT.

Static NAT Configuration.

* R1(config)# int g0/1
R1(config-if)# ip nat inside
→ Define the 'inside' interface(s) connected to internal network.

R1(config-if)# int g0/0
R1(config-if)# ip nat outside
→ Define the 'outside' interface(s) connected to external network.

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R1(config)# ip nat inside source static 192.168.0.167 100.0.0.1
R1(config)# ip nat inside source static 192.168.0.168 100.0.0.2

Configure the one-~~one~~-to-one IP add mappings.
ip nat inside source static inside-local-ip inside-global-ip

R1 # show ip nat translations

	Protocol	Inside-global	Inside-local	Outside-local	Outside-global
dynamic	udp	10.0.0.1:56310	192.168.0.167:56310	8.8.8.8:53	8.8.8.8:53
static →	---	10.0.0.1	192.168.0.167	---	---

Unless dst is used the two add. will be same

NOB: Remember port-no. doesn't get translated.

* Outside Local - The IP add. of outside host, from perspective of local network.

* Outside Global - The IP add. of outside host, from the perspective of the outside network.

Inside/Outside = Location of host
Local/Global = Perspective.

Few more useful commands.

R1# clear ip nat translation *
clears all the dynamic translation in NAT translation table.

* Each time the static NAT entries are ~~used~~ actually used, dynamic entries are added to the NAT translation table for those translations

* Static NAT entries ~~won't~~ can't timeout & ~~won't~~ be deleted unless we remove ip nat inside source static commands from router.

R1 # show ip nat statistics
Total active translations: 2 (2 static, 0 dynamic; extended)
peak-translation: 4 (highest translation in NAT table)

Day-45

June-14

NAT (PART-II) - Dynamic NAT

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